RAGHAVENDER REDDY BADUGULA

MSc Advanced Mechanical Engineering (Cranfield University) | Simulation (CFD/FEA), Materials & Project Engineering

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PROFESSIONAL SUMMARY

Mechanical, Simulation and Materials Engineering graduate (MSc, Cranfield University) with practical expertise in Wire Arc Additive Manufacturing (WAAM) high-temperature molten salt corrosion, Abaqus FEA structural buckling, ANSYS ICEM CFD and Fluent aerodynamic optimization, MATLAB and XFOIL Blade Element Momentum (BEM) and offshore wave mechanics, structural integrity, and engineering project management. Qualified in GATE 2022 (Smart Manufacturing) with six years of freelance technical localization experience (2018 to 2024), including serving as an Assistant Project Manager for confidential national-scale examination operations, alongside current UK frontline team leadership. Combines first-principles physical engineering analysis with AI-assisted software product architecture to design system logic and build end-to-end engineering and localization tools.

CORE TECHNICAL & MANAGEMENT SKILLS

- **CFD, FEA & Numerical Simulation:** Abaqus FEA (3D Shell Partitioning, Linear Perturbation and Eigenvalue Buckling), ANSYS Fluent (k-omega SST Viscous Turbulence Modeling), ANSYS ICEM CFD (Multi-Block Structured O/C-Grid Meshing, y+ Boundary Layer Sizing, Grid Independence), XFOIL, MATLAB (Blade Element Momentum [BEM] Solver, Airy Linear and Non-Linear Offshore Wave Kinematics), SolidWorks (3D CAD and Kinematics), AutoCAD.
- **Materials, Corrosion & Structural Integrity:** Wire Arc Additive Manufacturing (WAAM SS316L), High-Temperature Molten Salt Corrosion (600°C Ternary NaNO3/KNO3/KCl), 3D Optical Surface Profilometry and Dimensional Metrology (30-Position Stitched ZVI Interferometry, Cumulative Probability Analysis), SEM-EDS Elemental Mapping and Point-ID, Metallographic Sample Preparation, Precision Gravimetry (10^-5 g), Marine Coating Hypothesis Testing, Fracture Mechanics (FAD Kr/Lr, Stress Intensity Factor K, Paris-Law Fatigue Crack Growth), COSHH Safety Compliance.
- **Engineering Design & Project Management:** Project Charter and Scope Governance, Double Diamond Design Thinking, Quantitative PM Risk Assessment Matrices (Probability-Impact Scoring and Mitigation), Multi-Phase Gantt Scheduling, Circular Economy and UK/ISO Standards Compliance (BS EN 12464-1, ISO 3382-3, BS EN 1729-1, BS 8300), Rapid 3D Printing (Bambu Studio PLA) and Mechatronic Prototyping.
- **Product Logic & AI-Augmented Software:** End-to-End System Architecture and Workflow Logic Design, AI-Assisted Software Prototyping and Execution (Python, FastAPI, Flask, Playwright, Git/GitHub Actions CI/CD), Automated QA Pipelines, Offline and Cloud LLM Integration.

EDUCATION & QUALIFICATIONS

MSc in Advanced Mechanical Engineering | Cranfield University, UK — Oct 2024 to Sep 2025

- Key Modules: Fluid Mechanics and Loading (74%), Component Design (73%), Structural Integrity (69%), Industrial Group Project with Nissan (66%), Principles of Engineering (64%), Engineering Design and Project Management (63%), MSc Thesis (62%), CFD for Renewable Energy (60%), Engineering Stress Analysis (Abaqus FEA), Assessing Risk and Failure.

B.Tech in Mechanical Engineering (Mechatronics) | MGIT, Hyderabad, India — Aug 2016 to Nov 2020

- GATE 2022 Qualified: Smart Manufacturing and Advanced Production Systems. Core Focus: Mechatronics, CAD/CAM, Robotics Kinematics, and Thermo-Fluid Systems.

ENGINEERING, SIMULATION & DESIGN PROJECTS (CRANFIELD UNIVERSITY & MGIT)

Molten Salt Corrosion of WAAM SS316L for CSP Thermal Energy Storage (MSc Thesis) | Cranfield University — 2024 to 2025

- Subjected Wire Arc Additively Manufactured (WAAM) SS316L specimens to ternary chloride molten salts (10NaNO3-78KNO3-12KCl) at 600°C for 250 h and 500 h in a horizontal tube furnace under controlled nitrogen purge, comparing open crucibles against covered-lid crucibles to simulate vapour-retentive CSP storage tank zones.
- Prepared metallographic cross-sections through precision abrasive sectioning, cold-resin mounting, sequential SiC grinding, diamond polishing, and ultrasonic cleaning with acetone and isopropanol (IPA).
- Conducted 30-position stitched 3D optical dimensional metrology (.zvi surface profilometry and cumulative probability radius-change distributions), precision mass change metrology (10^-5 g accuracy), and cross-sectional SEM-EDS elemental mapping and Point-ID spectroscopy to evaluate Cr/Fe/Ni diffusion, Na/K/Cl/O ingress, and oxide scale spallation.
- Demonstrated that covered-lid vapour retention accelerates porous scale formation and localized pit initiation at 250 h compared to compact Cr/Fe-rich oxides in open crucibles, and authored complete COSHH risk assessments and high-temperature chemical safety protocols.

Fluid Mechanics & Offshore Loading: HAWT BEM Rotor & Airy Wave Kinematics | Cranfield University (74%) — Jan 2025

- Wind Turbine Rotor Aerodynamics (Part A): Generated NACA 0030 airfoil lift and drag polars in XFOIL and wrote an iterative Blade Element Momentum (BEM) convergence algorithm in MATLAB for a 3-bladed Horizontal Axis Wind Turbine (R = 7 m), computing radial induction factors, total thrust (1,804.6 N), torque (1,693.8 Nm), mechanical power (10.16 kW), and power coefficient (Cp = 0.1872), alongside blade pitch and twist optimization.
- Offshore Hydrodynamic Wave Loading (Part B): Applied Linear (Airy) and Non-Linear wave theory (H = 0.6 m, T = 10 s, wavelength = 43.69 m, depth = 2 m) to derive horizontal and vertical fluid velocity potentials, simulating depth-dependent velocity profiles and elliptical water particle trajectories (z = 0 to -1.9 m) in MATLAB to evaluate hydrodynamic wave forces on offshore monopile meteorological masts.

Engineering Stress Analysis: FEA Buckling of Offshore Wind Turbine Monopiles | Cranfield University — Abaqus FEA

- Modeled a 40 m offshore wind turbine hollow steel monopile foundation (outer radius R = 3.75 m, wall thickness t = 0.08 m) in Abaqus using 3D shell elements partitioned via horizontal depth planes (10 m atmospheric, 20 m seawater, 10 m seabed) and 90° circumferential datum planes with fixed encastre seabed boundary conditions.
- Constructed a 23,200-element structured rectangular mesh (0.2 global seed) to eliminate tetrahedral discretization noise and executed Linear Perturbation (Eigenvalue Buckling) analysis under combined eccentric rotor/nacelle weight, marine wave pressure (1.5 kPa), and sudden seabed pressure (20.3 kPa).
- Extracted and evaluated the first 5 critical buckling modes (eigenvalues lambda_1 = 0.3349 to lambda_5 = 0.5917), characterizing progressive multi-wave shell instabilities and fixed-base stress concentrations against DNV offshore structural standards.

ICEM CFD Aerodynamic Simulation & Biomimetic Airfoil Optimization for Urban VAWTs | Cranfield University ANSYS ICEM & Fluent

- Modified a symmetric NACA 0015 airfoil for urban Vertical Axis Wind Turbines (VAWTs) operating at low-to-moderate wind speeds (4 m/s and 12 m/s) by introducing a 40%-chord upper-surface bump and a stingray-inspired curved trailing edge to delay dynamic stall.
- Built a C-shaped computational domain (-20c upstream to +40c downstream) with multi-block O-grid structured hexahedral meshing in ANSYS ICEM CFD (first-layer wall spacing 3.3x10^-6 m for target y+, 2x2x2 determinant quality > 0.837), validated across a 4-stage grid independence study from 21,274 to 104,274 nodes (<1% variation).
- Simulated 2D incompressible viscous flows using the k-omega SST turbulence model in ANSYS Fluent, demonstrating prolonged upper-surface boundary layer attachment and increasing lift coefficient (CL) from 0.80 to 0.95 at 10° angle of attack.

Ship Hull Marine Corrosion & Paint Layer Hypothesis Testing (Assessing Risk & Failure) & Structural Integrity | Cranfield University 2024 to 2025

- Ship Hull Paint Layer Hypothesis Testing (Assessing Risk & Failure): Designed and executed controlled marine corrosion hypothesis testing and risk/failure assessment comparing single, double, and triple ship hull coating thicknesses; demonstrated empirically through statistical hypothesis testing that double-layer paint application significantly optimizes anti-corrosion protection efficiency and failure resistance over single-layer coating while avoiding triple-layer diminishing returns.
- Structural Integrity & Fracture Mechanics Assessment (69%): Evaluated linear-elastic and elastic-plastic fracture mechanics across cracked metallic structures, computing stress intensity factors (K), Failure Assessment Diagrams (FAD Kr/Lr interaction loci), and Paris-law fatigue crack propagation rates (da/dN vs. Delta K across Regions I, II, and III) under cyclic loading.

Chemical Preparation of a Novel Semiconductor Solar Paint | Group Project with Nissan, Cranfield University (66%) 2024 to 2025

- Formulated a lead-free photovoltaic solar paint matrix by dispersing bismuth methyl iodide quantum dots (1.8 to 2.3 eV tunable visible bandgap) within a TiO2 electron-transport semiconductor matrix via controlled mechanical and magnetic stirring.
- Prepared abraded galvanized automotive steel substrates and performed electrical conductivity and phase-stability testing using calibrated multimeters and optical equipment over 48-hour cycles, achieving a 5x conductivity enhancement in double-concentration samples while coordinating PhD lab mentorship and group technical report synthesis.

Smart & Sustainable Shower System: Component Design & 3D-Printed Prototype | Cranfield University (73%) SolidWorks & 3D Print

- Designed, modeled in SolidWorks 3D CAD, and 3D-printed (Bambu Studio PLA) a closed-loop household shower cold-water recirculation prototype integrating temperature sensors, solenoid directional valves, an under-tub buffer reservoir, and a return pump to the boiler.
- Conducted stakeholder influence analysis, functional and non-functional systems requirements definition, and physical studio prototype testing targeting a 50% to 70% reduction in domestic pre-heating water wastage.

Engineering Design & Project Management: Stafford Cripps SC3 Smart Classroom Modernization | Cranfield University Project Lead

- Authored the Project Charter, Double Diamond design-thinking justification, circular-economy furniture reuse plan, and UK/ISO regulatory compliance specification (BS EN 12464-1 LED lighting, ISO 3382-3 acoustic treatment, BS EN 1729-1 ergonomics, BS 8300 wheelchair accessibility).
- Constructed a 15-item quantitative PM Risk Assessment Matrix (reducing high-risk Probability-Impact scores from 25 to <= 9 via early procurement, UPS backup, and off-hour phased installation), Quality Assessment Matrix, and 11-week multi-phase Gantt schedule.

Multi-Terrain Quadruped Robot & Thermoacoustic Engine Simulation | Undergraduate Projects, MGIT 2016 to 2020

- Designed 3D quadruped mechanical leg assemblies and chassis structures using SolidWorks and evaluated walking gait kinematics and multi-terrain stability dynamics.
- Conducted literature research and thermal-acoustic wave simulation analysis for low-maintenance thermoacoustic energy conversion systems.

PRODUCT ARCHITECTURE & AI-AUGMENTED SOFTWARE SYSTEMS

Polyglot Engine: Autonomous Multilingual UI Layout Scanner & Repair CLI | github.com/Raghavender1234/polyglot-engine Product Architect (2026)

- Conceived and architected a 6-layer multilingual UI verification tool connecting live browser DOM geometry (Playwright), OpenType text shaping (HarfBuzz), and Unicode BiDi isolation to diagnose and repair translation overflow and CSS layout defects.
- Authored the end-to-end diagnostic and backtracking repair logic and directed AI-assisted software development workflows to implement, test (pytest and 50-case visual benchmark), and package a turnkey CLI and GitHub Action.

IndiCAT Workstation (Microsoft Store / indicat.co.in) & OpenLMS / PR DIA (github.com/Raghavender1234/pr-dia) Founder & Architect (2024 to Present)

- IndiCAT Workstation (Live on Microsoft Store): Conceived and architected a multi-format CAT and localization workstation (DOCX, Excel, Trados SDLXLIFF, Subtitles, and PDF across 22 Indian and 200+ global languages) featuring 100% offline IndicTrans2/NLLB and Cloud AI modes, Whisper voice dubbing studio, WhatsApp/Telegram bot sync, and built-in Translator CRM and invoicing.
- OpenLMS / PR DIA & UK Rota Optimizer: Designed the system logic and multi-tenant architecture for an AI-native Learning and Assessment platform (featuring a complete UPSC ESE Mechanical Engineering curriculum, Socratic AI Tutor, and timed exam simulator) alongside a constraint-based UK retail shift rota optimizer built with AI-assisted coding workflows.

WORK & OPERATIONAL EXPERIENCE

Team Lead / Operational Specialist | Smashburger, Milton Keynes, UK Nov 2024 to Present

- Promoted to Team Lead based on frontline operational performance; directing shift execution, staff workflows, quality control, food safety auditing, and inventory management in a high-volume UK hospitality environment.

Freelance Technical Translator & Assistant Project Manager (<1 Year) | Enterprise Assessment & Media Localization 2018 to Sep 2024

- Freelance Technical Translator & Localization Specialist (2018 to 2024): Delivered high-accuracy English-to-Telugu technical translation, subtitle localization for major streaming platforms (Netflix and Amazon Prime releases), and remote interpretation for UK corporate and legal clients.
- Assistant Project Manager / Operational Coordinator (2023 to 2024, <1 Year): Coordinated end-to-end multilingual translation, SME validation, and data workflows for confidential national-scale competitive examination and assessment projects; managed cross-functional freelance translator and typist teams under strict SLA deadlines and security protocols.

Industrial Engineering Internships | RINL Vizag Steel Plant & Maruti Suzuki (Varun Motors), India 2018 to 2020

- Completed industrial manufacturing and maintenance training at Rashtriya Ispat Nigam Limited (RINL Vizag Steel Plant: Coke Plant and Medium Merchant & Structural Mill I-beam rolling) and Maruti Suzuki automotive service operations.